Lesson 13 More Combinatorial Proofs

$\binom{0}{0}$

$\binom{1}{0}$ $\binom{1}{1}$

$\binom{2}{0}$ $\binom{2}{1}$ $\binom{2}{2}$

$$
\begin{array}{ccccccccc}
 & & & & 1 & & & & \\
 & & & 1 & & 1 & & & \\
 & & 1 & & 2 & & 1 & & \\
 & 1 & & 3 & & 3 & & 1 & \\
1 & & 4 & & 6 & & 4 & & 1
\end{array}
$$

$\binom{3}{1} = \binom{3}{2}$

$\binom{4}{1} = \binom{4}{3}$

Prop 1 $\binom{n}{k} = \binom{n}{n-k}$

Algebraic proof

$$\binom{n}{k} = \frac{n!}{(n-k)!\,k!} \qquad \binom{n}{n-k} = \frac{n!}{\underbrace{(n-(n-k))!}_{k!}\,(n-k)!}$$

Combinatorial proof

To construct a k-element subset of an n-element set, we may choose k elements to include OR we may choose n-k elements to exclude.

Therefore there are $\binom{n}{k}$ such subsets or $\binom{n}{n-k}$ such subsets, and $\binom{n}{k} = \binom{n}{n-k}$.

Prop 2 $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$

Pascal's Identity

Combinatorial Proof

LHS: # of k-element subsets of $A = \{1, 2, \ldots, n\}$

RHS: There are 2 types of such subsets:

(1) Subsets with "n": $\binom{n-1}{k-1}$

$\{n, \underbrace{\ldots\ldots}_{k-1}\}$

(2) Subsets without "n": $\binom{n-1}{k}$

$\{\underbrace{\ldots\ldots\ldots}_{k}\}$

Prop 3 $k\binom{n}{k} = n\binom{n-1}{k-1}$

Proof How many ways to choose from n people a team of k members with a captain?

We may choose the k team members first: $\binom{n}{k}$ and then choose the captain: $\binom{k}{1} = k$ (LHS)

We may choose the captain first: $\binom{n}{1} = n$, then choose the rest of the team: $\binom{n-1}{k-1}$ (RHS)

<u>Prop 4</u> $\binom{n}{2} = 1+2+3+\ldots+(n-1)$

<u>Proof</u> LHS: # of 2-element subsets of $A = \{1, 2, \ldots, n\}$

RHS: Here is a possible breakdown of the subsets:

There are <u>$n-1$</u> subsets with "1":

$\{1,2\}, \{1,3\}, \ldots, \{1,n\}$

There are <u>$n-2$</u> subsets without "1" but with "2":

$\{2,3\}, \{2,4\}, \ldots, \{2,n\}$

There are <u>$n-3$</u> subsets w/o "1" or "2" but with "3": $\{3,4\}, \{3,5\}, \ldots, \{3,n\}$

$\vdots$

There is <u>1</u> subset without "1" ... "$n-2$" but w: "$n-1$": $\{n-1, n\}$

Prop 5 $$\sum_{k=0}^{\ell} \binom{n}{k}\binom{m}{\ell-k} = \underbrace{\binom{n+m}{\ell}}_{\text{choose } \ell \text{ marbles from } n \text{ red marbles and } m \text{ blue marbles}}$$

$$\underbrace{\binom{n}{0}\binom{m}{\ell}}_{k=0} + \underbrace{\binom{n}{1}\binom{m}{\ell-1}}_{k=1} + \cdots + \underbrace{\binom{n}{\ell}\binom{m}{0}}_{k=\ell}$$

0 red + ℓ blue

1 red + $\ell-1$ blue

ℓ red + 0 blue